

HW6

5 ноября 2025 г. 16:27

Problem 1

23. How many elements does each of these sets have where a and b are distinct elements?

- a)** $\mathcal{P}(\{a, b, \{a, b\}\})$
- b)** $\mathcal{P}(\{\emptyset, a, \{a\}, \{\{a\}\}\})$
- c)** $\mathcal{P}(\mathcal{P}(\emptyset))$

A) $2^3 = 8$

B) $2^4 = 16$

c) $2^1 = 2$

Problem 2

37. How many different elements does A^n have when A has m elements and n is a positive integer?

$$|A^n| = m^n$$

$$m^n$$

Problem 3

43. Find the truth set of each of these predicates where the domain is the set of integers.

a) $P(x): x^2 < 3$

b) $Q(x): x^2 > 2$

c) $R(x): 2x + 1 = 0$

$$a) \{x \in \mathbb{Z} \mid x^2 < 3\} = \{-1, 0, 1\}$$

$$-\sqrt{3} < x < \sqrt{3}$$

$$\sqrt{3} \approx 1,732$$

$$-\sqrt{3} \approx -1,732$$

$$b) \{x \in \mathbb{Z} \mid x^2 > 2\} = \{\dots, -3, -2, -1, 2, 3, 4, \dots\}$$

$$c) \{x \in \mathbb{Z} \mid 2x + 1 = 0\} = \emptyset$$

$$2x + 1 = 0$$

$$2x = -1$$

$$x = -\frac{1}{2}$$

$$x = -2$$

$$x = -0,5$$

no integers

Problem 4

***47.** Describe a procedure for listing all the subsets of a finite set.

1. $\{\emptyset\}$
2. $+p : \{\emptyset, \{p\}\}$
3. $+d : \{\emptyset, \{p\}, \{d\}, \{p,d\}\}$
4. $+q : \{\emptyset, \{p\}, \{d\}, \{p,d\}, \{q\}, \{p,q\}, \{d,q\}, \{p,d,q\}\}$

Problem 5

17. Show that if A , B , and C are sets, then $\overline{A \cap B \cap C} = \overline{A} \cup \overline{B} \cup \overline{C}$

a) by showing each side is a subset of the other side.

b) using a membership table.

$$a) \overline{A \cap B \cap C} = \bar{A} \cup \bar{B} \cup \bar{C}$$

$$x \in \overline{A \cap B \cap C}$$

$$x \notin A \cap B \cap C$$

$$x \notin A \text{ OR } x \notin B \text{ OR } x \notin C$$

$$x \in \bar{A} \text{ OR } x \in \bar{B} \text{ OR } x \in \bar{C}$$

$$\overline{A \cap B \cap C} \subseteq \bar{A} \cup \bar{B} \cup \bar{C}$$

b)

A	B	C
1	1	1
1	1	0
1	0	1
1	0	0
0	1	1
0	1	0

$A \cap B \cap C$
1
0
0
0
0
0

$\overline{A \cap B \cap C}$
0
1
1
1
1
1

$\bar{A}$	$\bar{B}$	$\bar{C}$
0	0	0
0	0	1
0	1	0
0	1	1
1	0	0
1	0	1
1	1	0

$\bar{A} \cup \bar{B} \cup \bar{C}$
0
1
1
1
1
1

0

0

b

0

1

1

1

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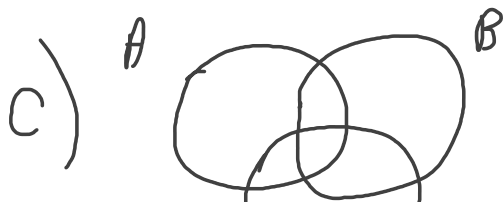
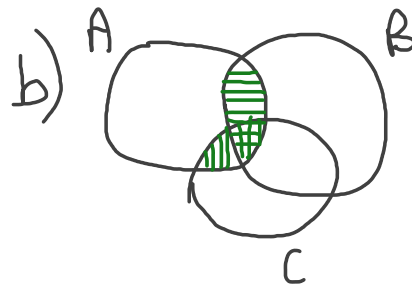
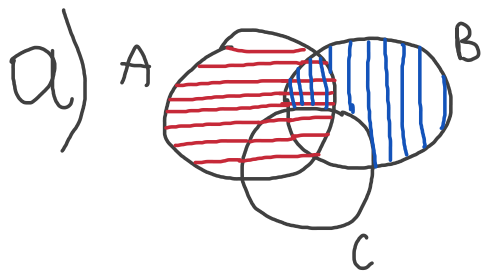
Problem 6

27. Draw the Venn diagrams for each of these combinations of the sets A , B , and C .

a) $A \cap (B - C)$

b) $(A \cap B) \cup (A \cap C)$

c) $(A \cap \overline{B}) \cup (A \cap \overline{C})$



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